Alpha Design Worksheet

Template for Alpha Design Assignment / Exam

Metadata

Alpha name	Author	Date

Universe	Neutralization	Decay	Truncation

Alpha Design Steps

Step	Guiding prompts	Your answer
1	**Define objective & edge** What market inefficiency do you believe exists? Why has it not been fully arbitraged? What outcome do you expect (e.g. mean revers	
2	**Formulate hypothesis** If X changes, what should happen to returns? Clearly define Long and Short positions. What is the expected holding horizon?	
3	**Mathematical expression** List all raw variables used. Define units, scaling, and data treatment. Write the base formula before transformations.	
4	**Transform & refine** Choose operators: z-score, rank, group-rank, n Specify lookback windows, decay, truncation. Expected direction: high = long or short?	
5	**Robustness & final alpha** Key risks or failure cases. Suggested robustness checks (subperiods, sec Final alpha formula (pseudo-code) and rational	

Worked Example: Mean Reversion Alpha

Illustrative example

1. Objective & Edge

Short-term price dislocations arise from liquidity shocks and investor overreaction.
These effects are temporary and tend to mean-revert.

2. Hypothesis

Stocks with extreme negative short-term returns will outperform,
while extreme positive return stocks will underperform.
Expected holding horizon: 2–10 trading days.

3. Mathematical Expression

Raw variable: daily returns.
Compute cumulative returns over a short window (e.g. 5 days).

4. Transform & Refine

Apply cross-sectional z-score to identify extremes.
Rank within industry to control sector exposure.
Low z-score = Long, High z-score = Short.

5. Final Alpha Formula

Pseudo-code:

```
group_rank( -zscore(ts_sum(returns, 5)), industry )
```

Rationale: exploits short-term mean reversion while neutralizing industry effects.

Alpha Design – Multi-Alpha Work Table

Each row = one alpha idea

Alpha	[redacted]	[redacted]	[redacted]	[redacted]	[redacted]

Guidance: each row is a distinct alpha; keep steps consistent across columns.

FINAL